

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the
Interstate Transmission System

Docket No. RM26-4-000

**INITIAL COMMENTS OF
THE CALIFORNIA PUBLIC UTILITIES COMMISSION**

The California Public Utilities Commission and the People of the State of California (“CPUC”) submit these Initial Comments in response to the Federal Energy Regulatory Commission’s (“FERC” or “Commission”) October 27, 2025, Notice Inviting Comments on the Advance Notice of Proposed Rulemaking (“ANOPR”) on large load interconnection.¹

The CPUC strongly supports the shared state and federal goal of ensuring prompt and efficient energization of large load resources, such as AI data centers. However, the CPUC objects to the Secretary of Energy’s proposition that the proposed rules do not venture into state jurisdiction. The Federal Power Act (“FPA”) bars the Commission from regulating retail sales, and the Supreme Court has been clear: specification of terms of sale at retail “is a job for the States alone.”² States have the expertise, authority, and ongoing and developed proceedings over areas impacted by this rulemaking: energizing

¹ To distinguish load interconnection from generator interconnection, California refers to interconnection of load as “energization.” Use of the term energization in these Initial Comments, unless otherwise stated, is synonymous with “load interconnection.”

² *Tri-State Generation and Transmission Association, Inc., Order Rejecting Proposed Tariff*, 193 FERC ¶ 61,070 (citing *FERC v. Elec. Power Supply Ass’n*, 577 U.S. 260, 264-65 (2016) (“*EPSA*”)).

retail load, retail rates and terms of service, integrated resource planning, greenhouse gas reduction policy, siting, and system reliability.

The CPUC and other states are already working on state-jurisdictional tariffs and standardized rules for large load energization. The CPUC is currently reviewing a utility application to establish a new tariff to streamline the energization process for transmission-level retail electric service customers, particularly from data centers, and a CPUC decision is expected mid-2026.³ States, including California, are already underway in addressing the concerns the Department of Energy (“DOE”) identifies, making broad federal intervention duplicative and counter to the goal of timely energization of load. It would be most efficient and constructive for the rulemaking to divide areas of responsibility and carve out the work already done by the states. The CPUC urges FERC to continue the tradition of cooperative federalism enshrined by the Federal Power Act and coordinate state and federal action for engineering-driven and efficient outcomes necessary to reach our mutual goals.

I. CONGRESS HAS IMBUED BOTH FERC AND THE STATES WITH AREAS OF JURISDICTIONAL AUTHORITY UNDER THE FEDERAL POWER ACT

In passing the Federal Power Act and the amendments in the Energy Policy Act of 2005 (“FPA”), the U.S. Congress has ensured a system of cooperative federalism and given complementary jurisdictional powers to federal agencies and the states. Under the

³ A.24-11-007, *Application of Pacific Gas and Electric Company (U39E) for Approval of Electric Rule No. 30 for Transmission-Level Retail Electric Service* (Nov. 21, 2024) (“Rule 30 Application”). Interim implementation was approved on July 24, 2025 in CPUC Decision 25-07-039.

FPA, FERC has jurisdiction over the transmission of electric energy in interstate commerce, including the rates, terms, and conditions of transmission service, and the “sale of electric energy at wholesale rates in interstate commerce,” and any rule or practice affecting such rates.⁴ Further, the FPA limits federal regulation “to extend only to those matters which are not subject to regulation by the States.”⁵ Under established rules of statutory interpretation, by being specific on the jurisdiction given to FERC and the States, Congress did not give FERC exclusive jurisdiction over all things tangential to the transmission system or that have incidental impact on wholesale electric rates. Instead, Congress gave jurisdictional authority to both FERC and the states and did not preempt the whole field of electric transmission. As articulated by the United States Supreme Court, FERC has authority under the FPA to ensure that practices *directly* affecting wholesale rates are just and reasonable⁶ – however, one of the limits on FERC’s authority is that state utility commissions have jurisdiction over retail transactions.⁷

Large load interconnection is part of retail service and is not part of the transmission system operation or a purchase of wholesale electricity. Large load customers are end-use retail load customers, and their interconnection and purchase of electricity is a retail transaction, the terms and conditions of which are subject to state jurisdiction.

⁴ *FERC v. Elec. Power Supply Ass’n*, 577 U.S. 260, 264-65 (2016) (“*EPSA*”) (citing 16 U.S.C. §§ 824(b), 824e(a)).

⁵ 16 U.S. Code § 824.

⁶ *Id.* at 279-80. (emphasis added).

⁷ *Id.* at 267.

The following sections address each of the ANOPR's four legal justifications for federal jurisdiction over retail large load interconnection.⁸

A. Generator Interconnections and Large Load Interconnections Are Not Analogous

The ANOPR identifies the Commission's existing Large Generator Interconnection Procedures and asserts that "like generator interconnections, large load interconnections are a critical component of open access transmission services that require minimum terms and conditions to ensure non-discriminatory transmission service."⁹ This analogy between generator and large-load interconnections is too simplistic and does not account for fundamental jurisdictional and technical distinctions established under the FPA.

1. Jurisdictional Differences Between Generation and Retail Load Merit Different Treatment

FERC's Open Access Transmission Tariff ("OATT") governs generator interconnection for the purpose of wholesale sales and transmission service with the goal of preventing discrimination and providing equal access to the transmission system.¹⁰ In contrast, the ANOPR would create rules for large load interconnection – the purpose of which is retail end user customer purchases. While large loads like AI data centers may

⁸ ANOPR at PP 13-16.

⁹ ANOPR at P 13.

¹⁰ Promoting Wholesale Competition Through Open Access Non-discriminatory Transmission Services by Public Utilities; Recovery of Stranded Costs by Public Utilities and Transmitting Utilities, Order No. 888, 61 Fed. Reg. 21,540 (May 10, 1996), FERC Stats. & Regs. ¶ 31,036 (1996), order on reh'g, Order No. 888-A, 62 Fed. Reg. 12,274 (March 14, 1997), FERC Stats. & Regs. ¶ 31,048 (1997), order on reh'g, Order No. 888-B, 81 FERC ¶ 61,248 (1997), order on reh'g, Order No. 888-C, 82 FERC ¶ 61,046 (1998), aff'd in relevant part sub nom. Transmission Access Policy Study Group v. FERC, 225 F.3d 667 (D.C. Cir. 2000), aff'd sub nom. New York v. FERC, 535 U.S. 1 (2002). ("Order 888")

interconnect directly to the transmission grid due to their voltage needs, and their high demand may have an impact on the transmission grid, their consumption remains retail load, that does not implicate purchasing wholesale electricity. As the US Supreme Court has stated, “no matter how direct, or dramatic” a proposal’s impact on wholesale rates, the Commission still may not regulate retail electricity sales.¹¹ By statute and precedent, states set the minimum terms and conditions of retail service. The OATT for generator interconnection does not raise the same concern of intruding into matters subject to the jurisdiction of the states,¹² as states do not regulate electric generation.

2. Technological Differences Between Generation and Retail Load Merit Different Considerations

The electric system is more than just a marketplace for the exchange of generation and load services; it is a physical network that requires carefully managed processes to guarantee safety and reliability. Interconnecting generation and interconnecting load have technological differences that result in different impacts on the transmission system and drive different considerations. While there are sound technological reasons for the Commission to regulate the delivery of electrical generation, the same rationales do not apply to electrical load.

To begin, generator and load interconnections differ fundamentally in their design and operational characteristics. Load interconnections currently do not require the same extensive control and protection equipment to manage injections into the system within

¹¹ *EPSA* at 279-80.

¹² 16 U.S. Code § 824.

strictly defined limits as generation interconnections do. Load interconnections also do not require generation controls, communication equipment, and protection schemes and equipment that are separate and distinct from equipment needed to connect generation to the system and which warrant distinct technical and procedural requirements. Loads, unlike generation resources, currently do not shoulder the responsibility of adjusting to reestablish system balance, voltage, and general stability.

Additionally, factors such as meter placement, and state and local permitting requirements (e.g., life, safety, electric, fire, etc.) further support separate treatment for generation and load interconnections.

Generator interconnections further differ from load interconnections because generation includes voltage support, establishing a healthy voltage profile throughout the grid in transferring power while also minimizing losses. The alternative to generation-related volt-ampere reactive (“VAR”) support is to deploy fixed-value capacitors and reactors throughout the grid, or to utilize much more costly VAR sources. Generator interconnections benefit from the cluster study process where multiple generators can be studied together and benefit from sharing the cost of certain network upgrade facilities, including new transmission lines, that can benefit multiple generators. Generator interconnections already require coordination and study with the local transmission owning utility and agreements are signed between the CAISO and the participating transmission utility. Only with the help of the local transmission utility are the generator specific network facilities identified and planned. Load interconnection presents technical requirements, distinct from those of generation, that render the state the only

appropriate jurisdictional authority. Large load withdrawals depend on being strategically located, sized, and prescribed for the range of unique safety and reliability challenges, a feature provided at generation sites through VAR support, ample short circuit strength, and inertia. VAR support and regional voltage profiles through a wide range of system conditions are most effectively planned and managed at the local level. Large loads are already factored into transmission planning at the system level and transmission planners coordinate with local transmission owners to plan reliability upgrades needed to serve the evolving load and generation landscape. Sometimes large loads require customer specific protection equipment to prevent grid disturbance from highly fluctuating loads, but this protection does not need to be studied collectively and is going to best be designed in coordination with the local transmission owner, who has closest visibility into the local engineering requirements of the system.

Large loads also form influential connections to local communities and become an important source of value-added tax funds, creating a greater interplay between local laws and regulations with large load corporations beyond energy connections established with generation facilities. State-wide and location-based grid considerations—such as siting and safety requirements—are best done through regulatory oversight at the state and local utility levels, where conditions on the ground are better understood.

B. The Process of Interconnecting Large Load Retail Customers Does Not Have a Direct Impact on Wholesale Transmission Rates

The ANOPR justifies its authority over the interconnection of large retail loads by asserting that “the interconnection of large loads to the transmission system falls under a practice directly affecting Commission-jurisdictional wholesale electricity rates.”¹³

While the large load entities’ *retail demand and purchases* of electricity – retail sales – will have an undeniable impact on wholesale transmission rates (and the economy writ large), the process of interconnection of large loads to the transmission system has no more of a direct effect on Commission-jurisdictional wholesale electricity rates than many other activities under the purview of states’ authority. Measures under the states’ authority, such as interconnecting smaller retail loads to the distribution system and possibly the transmission system, as well as safety and siting, could also be seen as having an effect on the wholesale electricity market.

The Commission accurately points out that “The FPA has vested the Commission with exclusive authority to ensure that wholesale rates are just and reasonable and not unduly discriminatory or preferential.”¹⁴ However, these are not wholesale rates – these are retail transactions to serve end use customers, and states are charged with ensuring the terms and conditions of retail service are just and reasonable.

¹³ ANOPR at P 14.

¹⁴ *Id.*

C. The Proposal Impacts Retail Rates, Terms, and Conditions Subject to State Authority

The ANOPR posits that “the proposal does not impinge on States' authority over retail electricity sales...” and “... does not exert jurisdiction over any retail sales to the large load.” Further, the ANOPR asserts that “nothing in the proposed reforms governs the siting, expansion, or modification of generation facilities.”¹⁵

To the contrary, asserting jurisdiction over the interconnection of large loads to the transmission system would impinge on states' exclusive authority over retail electricity sales. Interconnection of large load retail customers has a substantial effect on retail electricity sales – and the states have substantial concerns that AI load interconnection could drive up electric costs for other retail ratepayers, especially residential customers, if cost allocation is not properly considered at the state level. While the Commission asserts it is not seeking to directly regulate retail sales of electricity, the ANOPR would inevitably have the effect by influencing which retail loads may connect, at what times, and at what costs. Large load interconnection affects retail electric rates for all retail customers in a given state's jurisdiction, and therefore the state's interest and scope of authority.

Although the proposals in the ANOPR are framed as addressing only the timing and costs of interconnection upgrades; such upgrades are terms and conditions of retail service, the implications of which extend directly to retail rate design, driven by the state policy on cost allocation (between utility and customer) and cost recovery. The

¹⁵ ANOPR at P 15.

repayment of those upgrade costs is based on rates, specifically how soon utilities incur the costs to get the upgrades completed, allow the customer to interconnect when they pay for the upgrades, pay back the large load customer for upgrade costs if paid in advance by the customer and then add the system expansion investment costs to rate base, start depreciating the interconnection (*i.e.*, system expansion investments) facilities, and gain an authorized rate of return on those investments. The implementation of a retail sales tariff under which a customer obtains interconnection service for its large load facility has historically been delegated to the states because these large loads are end-use customers subject to retail rate structures that fund grid services and ensure cost allocation that is suitable to the local system conditions. To the extent that there are regional differences across the states in the timing, overall cost, and cost allocation rules applied to interconnection of large loads, they may provide a competitive signal for the efficient siting of new loads across various regions and is not necessarily a problem that requires intervention.

D. It Does Not Follow that Because the Commission Has Jurisdiction Over the Transmission System that Any Large Load Seeking to Interconnect to the Transmission System Falls Under Its Authority.

The ANOPR accurately points out that FERC has “exclusive jurisdiction over the transmission of electric energy in interstate commerce, including the rates, terms, and conditions of transmission service, and all facilities for such transmission or sale of electric energy at wholesale in interstate commerce.” However, the ANOPR takes a conceptual leap in asserting federal authority when it continues that “[a]ny large load that

seeks to interconnect to the transmission system does so to obtain transmission service and the appurtenant benefits of such.”

It simply does not follow that any large load entity seeking interconnection, and as a result receiving the benefits of the transmission system, falls within the sphere of the transmission regulation described above, particularly as most of these large load entities are retail end-use customers. Most retail end-use customers interconnect with the expectation that transmission service will be employed at some point to serve their load and receive benefits of interconnection to the transmission system.

Even when a large load facility takes service from the interstate transmission system, the physical location of the facility remains determinative of how it gets interconnected. Whereas FERC maintains jurisdiction over the assets determined to be part of the interstate transmission system per the Seven-factor¹⁶ and Mansfield¹⁷ Tests, states retain authority over the retail load customer interconnection process including siting studies, safety and code compliance inspections, and compliance with transmission owner regulations on interconnection, including interconnection studies. These state functions are essential to ensuring local reliability and public safety, and important reasons that load interconnection has been, and should continue to be, under states’ authority.

¹⁶ FERC Order 888 at P 401-402 (April 24, 1996).

¹⁷ *Mansfield*, 97 FERC ¶ 61,134 at 61,613-14 (November 7, 2001).

Finally, there appears to be inconsistency in how the ANOPR frames the legal justifications. The first legal justification reasons that large load interconnections are a component of open access transmission service, while the fourth legal justification states that large load interconnections obtain transmission service. This inconsistency blurs the line between transmission providers and end-use customers. A large load seeks to “obtain transmission service” and is not itself a component of the actual transmission service. Because a large load interconnecting to receive transmission service is not part of the open access transmission service itself, and is in fact distinct from generators accessing the transmission grid, oversight of interconnecting large retail load customers to the transmission grid should remain under the domain of the states.

II. COMMENTS ON THE ANOPR’S PRINCIPLES FOR REFORM

The CPUC aligns with DOE and FERC in the goal of “ensur[ing] all Americans and domestic industries have access to affordable, reliable, and secure electricity.”¹⁸ The CPUC similarly shares the goal of supporting American AI innovation through prompt and efficient connection of large loads to the transmission system.

The CPUC appreciates the opportunity to comment on this issue and provides the following responses to ANOPR’s Principles for Reform.

Principle 1. Federal-State Jurisdiction Balance

The CPUC appreciates the ANOPR’s acknowledgement of the areas in which the states have jurisdictional authority and perspectives on how large load interconnection

¹⁸ ANOPR, DOE Letter at 1 (October 23, 2025).

may fit into Congressional authority granted to FERC. However, for the reasons discussed in detail above, the CPUC believes that regulating the terms of large load interconnection, which are retail end-use customers, encroaches into areas that are under the states' authority and does not strike the best federal-state jurisdictional balance consistent with the FPA.

Principle 2. Definition of Large Load

The ANOPR proposes that the reforms should apply to new loads greater than 20 MW and for hybrid facilities, defined as hybrid load and generation, where the load is greater than 20 MW. The Commission “seeks comment on alternative thresholds, including whether such a threshold is necessary at all.”¹⁹

What constitutes a large load is context specific in a quickly changing environment. Load growth is expected to increase dramatically over the coming years, and what constitutes a “large” load will likely change over time. For implementable rules that do not require frequent revisions, states should retain flexibility of defining the retail large load class based on market and technical local system conditions.

The ANOPR's proposed 20 MW appears to have been proposed as an analogue to generator interconnection rather than through review of the existing and forecasted data center load complexion. The CPUC observes that data centers connecting to the transmission system in California are substantially larger than 20 MW, in some instances; recent filings have been more than four times that level with expected future filings being

¹⁹ ANOPR at P 19.

much larger than that. The CPUC is concerned that a threshold as low as 20 MW for large load energization will ensnare small commercial and industrial load in a large load energization process, which is not the intended goal of the ANOPR. Therefore, while reiterating that there is no appropriate threshold for federal oversight of retail large load energizations, if a threshold is established, the CPUC believes that it should be far greater than 20 MW. If the Commission used the generator analogy, it would not be the size of the load itself but rather the classification of the line that serves as first point of interconnection that determines whether it is distribution or transmission – that would govern the jurisdiction.

Further, while large-load energizations share some operational similarities with generation interconnections, the two serve fundamentally different functions on the grid and therefore require distinct technical and procedural treatment. These two distinct types of connections to the grid should not be treated the same because of the unique requirements for injection and withdrawal and the unique cost and responsibilities inherent in these processes. Each connection type requires its own set of protections and facilities to maintain safety and reliability of the transmission system.

Principle 3. Studies of Load and Generating Hybrid Facilities

The ANOPR reasons that large load and hybrid facilities should be studied together with generation, which will result in more efficient siting of loads and generation, and minimize the need for costly network upgrades.²⁰

²⁰ ANOPR at P 20.

While it may be efficient and potentially less costly to site large load near or at the same point of interconnection as a new generating facility, states are in a better position to consider local conditions and have established jurisdiction over siting. As highlighted previously, the different requirements for injection and withdrawal, and the different cost and responsibilities inherent in these processes, support the conclusion that energization of load and interconnection of generation are best considered independently and separately. To the extent that hybrid facilities apply for interconnection – there is not necessarily a new problem that needs to be solved: generators with some native load have long been studied as generation, and load with some onsite generation has long been studied for connection as load. For example, solar generation with onsite storage has a different profile for studying in a generator interconnection queue than solar generation alone. Likewise, a large commercial load with onsite, behind the meter solar generation has a different profile than one without any generation.

Specific to California, adding this new layer of study processes would impact CAISO's recent interconnection process enhancements, approved by FERC last year,²¹ and their interaction with the CAISO Transmission Planning Process (TPP). For example, one of the enhancements includes the ability for load serving entities to “score,” and thus impact the prioritization of, generation and storage projects that are studied. If any Final Rule in this FERC docket changes that prioritization process by prioritizing large load studies, it would also impinge on established state planning and procurement

²¹ On November 7, 2025, FERC upheld all aspects of the CAISO's interconnection process enhancements in its Order Addressing Arguments on Rehearing in FERC Docket No. ER24-2671-001.

processes, potentially delaying projects needed to comply with procurement orders and maintain system reliability.

Principle 4. Study Deposits, Readiness Requirements, and Withdrawal Penalties

The ANOPR suggests that “like generating facilities, load and hybrid facilities should be subject to standardized study deposits, readiness requirements, and withdrawal penalties.” The Commission seeks comment on “the extent to which the existing study deposits, readiness requirements, and withdrawal penalties can be adopted.”²²

The CPUC supports ratepayer protections and efforts to mitigate risks associated with stranded or mis-sized transmission assets. In coordination with parties and utility representatives, the CPUC is currently evaluating standardized mechanisms for energization of large load customers, and is weighing the benefits of financing, penalties for the failure of load to materialize, and requirements for refund issuances.

The CAISO generator interconnection process incorporates provisions for deposits, readiness requirements, and penalties. CPUC’s energization rules provide similar guardrails, and the CPUC is considering modifications to state processes for large load connections in new or existing rules. The CPUC looks forward to continued collaboration with the CAISO through various pre-existing mechanisms, including our 2022 Memorandum of Understanding,²³ to ensure coordination and share learnings to

²² ANOPR at P 21.

²³ ISO CEC and CPUC Memorandum of Understanding - Dec 2022 (“2022 Memorandum of Understanding”).

enable effective state planning and timely energization of large loads and interconnection of the necessary generation to power such loads.

Principle 5. Hybrid Facilities Should Be Studied Based on Both Injection and Withdrawal Needs.

The ANOPR proposes that hybrid facilities be studied based on the amount of injection and/or withdrawal rights requested, claiming that this would incentivize co-location with new generation facilities and ensure efficient buildout of the transmission system.²⁴

The CPUC supports studying hybrid facilities based on requested injection and requested withdrawal rights, respectively. This is distinct from studying the net of withdrawal plus injection. Taking the Principle's hypothetical large load customer who withdraws 500 MW and has a 600 MW generating facility as a springboard for discussion, the time(s) of day and whether and when the customer is injecting or withdrawing from the system are important considerations. Depending on the configuration, it could be that this large load hybrid facility is functioning as a 100 MW generator or that this customer is placing both a 500 MW withdrawal and 600 MW injection strain on the system, at different times of day. Technical requirements should account for that possibility. Moreover, the extremes of either a 500 MW load condition or a 600 MW generation injection condition could result from a forced outage of either the load or the generation: a scenario that would be unaccounted for if only studying net withdrawal plus injection. Consistent with its position regarding the ANOPR's Principle

²⁴ ANOPR at P 22.

#3, the CPUC harbors concerns that the Principle #5 proposal would impact CAISO's recently approved interconnection process enhancements (IPE), as well as the IPE's interaction with CAISO's Transmission Planning Process. The IPE employs load serving entities' ("LSE") study and "scoring" of generation and storage to help prioritize projects. Any rule changing the IPE by prioritizing large load studies and interconnection would impinge on state planning and procurement processes, potentially delaying projects needed to comply with CPUC orders.

Principle 6. System Protections for Hybrid Facilities to Prevent Unauthorized Injections or Withdrawals

The ANOPR seeks comment on operational limitations and whether there should be "minimum technical requirements for ... system protection facilities, whether a hybrid interconnection customer should be subject to penalties for unauthorized injections or withdrawals, how any such penalties should be designed, and how such penalties should be allocated to other transmission customers."²⁵

The CPUC supports grid planning that incorporates both injection and withdrawal levels. The scale of the large loads currently being contemplated amplifies grid planning's dependence on injection and withdrawal location, scope, and scale. The CPUC supports the collaborative development of standards for technical requirements through the North American Electric Reliability Corporation (NERC) and associated entities including ISOs and RTOs to effectively serve large loads safely and reliably.

²⁵ ANOPR at P 23.

Additionally, penalties or other appropriate guardrails to limit excesses that may introduce safety or reliability problems could be considered.

Principle 7. Expediting Interconnection of Curtailable and Dispatchable Loads

The ANOPR proposes that “the interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited.” It further seeks comment on “whether this should be accomplished through a serial interconnection study process or by some other means” and “whether such studies can be completed in 60 days.”²⁶

With the ANOPR’s proposal that the study of large loads be integrated into the study of generation in Principle #3, it is unclear if expediting the study of curtailable large loads would mean prioritizing them over the existing generator interconnection queue, which could have various negative impacts on existing planning and procurement processes.

While this ANOPR principle proposes the prioritization of studies of curtailable and dispatchable loads, the CPUC is just as interested in including commitments of large customers that are technically capable of curtailing or shifting load in transmission planning, generator interconnection studies, and resource adequacy planning. A recent Duke University brief, *How DOE’s Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility*²⁷ argues that the Transmission System Planner

²⁶ ANOPR at P 24.

²⁷ Farmer et al., *How DOE’s Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility*, November 2025 (“Duke Brief”). <https://hdl.handle.net/10161/33530>

and the Balancing Authority need to have a framework to account for any transaction curtailment agreements or flexible interconnection agreements with flexible large load customers as part of the planning processes. The Duke Brief observes that commitments do not currently translate into avoided network or capacity investments because transmission planners and reliability entities must still assume the customer will draw at full peak levels. This likely results in “overbuild” of transmission and generation capacity, undermining the cost and timing benefits that flexible large-load interconnections could provide.

While the Brief asserts a need for FERC to create a structure for large-load flexibility commitments, the CPUC believes that with existing state constructs, such as its Memorandum of Understanding with the California Energy Commission and the CAISO on transmission and resource planning,²⁸ and relevant proceedings at the state level, the efficiencies identified in the Duke Brief can be realized. For example, the CPUC recently issued rate design guidance to California's IOUs for dynamic retail rates in Decision (D.) 25-08-049. This guidance is reflected in Southern California Edison’s (“SCE”) pending large-power dynamic rate proposal (CPUC Application A.24-06-014), in which SCE proposes "two-part" retail tariffs that combine a subscription (i.e., a contracted level of service capacity) with a dynamic energy price that varies hourly with system conditions. This dual participation could be seen as a requirement of service: the customer’s firm obligation would extend only to its subscribed level, with higher usage subject to

²⁸ 2022 Memorandum of Understanding.

curtailment at the operator's direction and penalties for non-performance. This approach mirrors the contractually based "transactional flexibility" proposed in the Duke Brief. The CPUC contends that, like other elements proposed in the ANOPR, the benefits of curtailment and load shifting commitments can be realized at the state level without federal oversight of retail large load energizations.

Principle 8. Responsibility for Network Upgrades

The Commission recommends that load and hybrid facilities be responsible for 100 percent of the network upgrades that they are assigned through the interconnection studies, and seeks comment on whether such costs should be offset through a crediting mechanism and, if so, over how many years.²⁹

The CPUC has been weighing the issue of network upgrade cost responsibility in its "Rule 30" proceeding (PG&E Application A.24-11-007) and appreciates that the Commission included this important topic in the ANOPR. The CPUC supports cost allocation "in a manner that is at least roughly commensurate with estimated benefits."³⁰ For any cost offsets (i.e., reimbursements to large load customers) for transmission network upgrades, it is imperative that they occur over a period sufficient to ensure that risks to other ratepayers associated with the failure of generation to timely materialize, and costs borne while load ramps up, are fully mitigated. Depending on factors including, but not limited to, the nature of the large load facility, the specifications of the

²⁹ ANOPR at P 25.

³⁰ FERC Order No. 1000, *Transmission Planning and Cost Allocation by Transmission Owning and Operating Public Utilities*, 136 FERC ¶ 61,051 at PP 622, 637 (2011).

upgrade work needed, and the anticipated revenue from the large load customer, the appropriate crediting time may differ. Again, while the CPUC does not agree with the Commission asserting its jurisdiction here, if it were to proceed, the CPUC encourages the Commission to consider ratepayer protections against stranded network upgrade investments in instances where such investments would not have occurred but for the expectation of large load delivery.

In California, the CAISO's generator interconnection process employs a "participant financing" model, in which the CAISO assigns a customer the costs of network upgrades. While the costs are financed by the interconnecting customer, once the generator is online, the participating transmission owner pays the customer back for the costs of the network upgrades over a five-year period, up to a cap per MW of generation. The ANOPR proposes to treat load interconnection similarly, but fails to account for the substantially different value add of generator interconnection to the system. Unlike load energizations, reimbursement of network upgrade costs to a generator is justified by resource adequacy requirements. Further, the five years over which payback to generators occurs is likely too short a period to ensure that load will develop and be sustained. Any reimbursement to load customers for their upfront investment in load energization upgrade costs would likely need to occur over longer than five years to ensure protection of other ratepayers.³¹

³¹ In its Rule 30 Application, PG&E proposed a 10-year refund period at p. 9.

The interconnection study process is a critical piece of the cost puzzle. The CPUC supports robust participation from impacted state and regional entities and affected Stakeholders.

Principle 10. Generator System Support Resource / Reliability Must Run Study

The ANOPR proposes that an existing generating facility that seeks partial suspension to serve newly co-located load must go through a system support resource (SSR) or reliability must run (RMR) study, which must consider system conditions and forecasted load growth at least three years after the proposed suspension. The suspension could only occur after needed network upgrades for reliability were in service, and the network upgrades would be the responsibility of the generator. The Commission seeks comments on “whether and how resource adequacy should be considered in the SSR/RMR type study.”³²

The CPUC is supportive of considering reliability via the proposed studies. When a new large load customer comes online in co-location with existing generation, that generation is not subsumed. Rather, the resource would first have to be evaluated for grid system and / or local needs before it could reduce the capacity value it provides to the grid. California relies on the existing resources to meet our existing and growing demand needs and does not allow resources to retire if they are needed to meet these needs. If a new customer that was not adequately planned for connects to the system, it should not be allowed to subsume existing resources that are serving existing demand.

³² ANOPR at P 27.

Further, regarding the horizon of such studies, in California’s experience, studies looking out only three years would likely be insufficient to fully assess system conditions, load growth, and potential reliability impacts.

Principle 11. Transmission Service Based on Withdrawal Rights

The Commission’s eleventh principle for reform states that “utilities serving large loads, including those at hybrid facilities, should be responsible for transmission service based on their withdrawal rights, as that value amount reflects the quantity of capacity and energy that is being transmitted across the transmission system to the load.”³³

The CPUC’s analysis indicates that, to the extent transmission infrastructure is constructed for a hybrid large load, withdrawal rights could be optimized by utilizing peak shaving at opportune times to capture the benefits of more energy sales while minimizing the need for new or upgraded transmission infrastructure. The contingency condition in which the large load is forced off while the generation is injecting at full capacity would need to be safely and reliably mitigated.

The CPUC contends that the transmission system needs to operate reliably while the hybrid facility is operating in either full load or full generation mode. If the transmission system capacity near the hybrid large load is not inherently capable of safely and reliably responding to the forced outage of either the hybrid facility’s load or generation, a complex protection scheme must be developed and deployed in response to each negative impact possible.

³³ ANOPR at P 28.

Principle 13. Plan of Implementation

The Commission's thirteenth principle for reform states that "there must be a plan to implement these proposed reforms" including "transition plans" and "treatment of large load interconnections that are already being studied for interconnection."³⁴

Given the CPUC's concern that the proposals in this ANOPR impinge on states' jurisdiction over the energization of large loads and generation, at the very least, the Commission should defer to rulemakings already underway by state regulators and, further, provide the opportunity for states to implement rules and regulations in the future to ensure the timely and cost-effective energization of large load and hybrid facilities, with greater assurance these state rules will result in just and reasonable rates for ratepayers.

In the event that the Commission does exercise authority over large retail load energizations, the CPUC encourages the Commission to work closely with each state's regulator to ensure the utmost flexibility and consideration of local factors.

Principle 14. NERC Reliability Standards

The Commission's fourteenth principal for reform states that "utilities serving large loads must meet all applicable NERC reliability standards and OATT provisions. Utilities and we must be prepared to revise large load energization procedures and agreements, as necessary. NERC should review its reliability standards to determine if

³⁴ ANOPR at P 30.

new registration categories or new or modified reliability standards are required to ensure reliability of the BES.”³⁵

The CPUC agrees that NERC reliability principles must be a primary consideration when energizing new large loads. At risk is the ability for generation in one part of the region to help serve demand in another part of the region because the transfer capability will no longer be available to schedule and transfer that power over those long distances. Given that the original benefit of a wholesale generation market was the opportunity for a more optimized generation dispatch, such generation optimization could be weakened with diminished transmission transfer capability. This also follows from the loss of transfer capability and reduced wholesale generation optimization.

III. THE CPUC AGREES WITH THE RECENT RESOLUTION ON THE ANOPR ADOPTED BY THE NATIONAL ASSOCIATION OF REGULATORY UTILITY COMMISSIONERS (NARUC).

The CPUC expresses its support for, and alignment with, NARUC’s *Resolution Urging the Federal Energy Regulatory Commission to Preserve and Affirm State Retail Regulatory Jurisdiction in its Large Load Interconnection Proceeding*³⁶ (“Resolution”), which was approved by the NARUC Board of Directors on November 11, 2025. The NARUC Resolution “urges the Federal Energy Regulatory Commission to preserve and affirm States’ retail regulatory authority under the Federal Power Act, ensure that large

³⁵ ANOPR at P 31.

³⁶ National Association of Regulatory Utility Commissioners *Resolution Urging the Federal Energy Regulatory Commission to Preserve and Affirm State Retail Regulatory Jurisdiction in its Large Load Interconnection Proceeding* at pp. 6-7 (November 11, 2025). (“NARUC Resolution”)

load interconnections do not compromise grid reliability or impose undue costs on retail customers, and respect state tools for promoting system flexibility and equitable cost allocation.”³⁷

Among other points, the NARUC Resolution also:

- Urges that FERC “make clear that it is affirmatively not asserting jurisdiction over end-use sales, which falls squarely within the exclusive jurisdiction of state retail energy regulatory authorities;”³⁸
- Contends that states have authority over integrated resource planning, “enabling states to oversee utilities’ long-term forecasting of electricity demand and evaluation of supply- and demand-side resources to meet that demand in a cost-effective, reliable, and sustainable manner;”³⁹
- Asserts that “States retain the ability to implement protective measures, such as large load tariffs and flexibility requirements, and the ability to incentivize large loads to invest in local communities, including through programs such as facilitating large loads to interconnect more rapidly while bringing direct economic benefits to the community in the form of funding virtual power plants, non-transmission alternatives or demand-side management programs;”⁴⁰
- Recognizes that “state regulatory authorities have the expertise and authority to balance economic development associated with large loads with the protection of retail customers, and seek to ensure the ANOPR proceeding does not undermine these efforts, disrupt orderly grid planning, and lead to further resource adequacy and affordability challenges;”⁴¹ and
- Observes that “at least 20 States have approved or have pending large load tariffs or similar measures, which may include financial commitments, curtailment protocols, and minimum contract terms to allow for the rapid interconnection of large loads without

³⁷ *Id.* at p. 7.

³⁸ *Id.* at p. 6.

³⁹ *Id.*

⁴⁰ *Id.* at p. 7.

⁴¹ *Id.*

compromising grid reliability or unduly burdening existing retail customers.”⁴²

IV. CONCLUSION

The transmission system enables a robust wholesale generation market, the foundation of which is adequate and reliable transfer capability, allowing diverse generators in a wide region to competitively bid to serve loads. This generation optimization to the lowest-cost generation in the market was a key driver of FERC’s Order 888.

Generation interconnections and load energizations are critically distinct. Generation is the product being sold into the energy market, which is sought for use by the large loads. As the product being sold, the producer takes responsibility for both the voltage and frequency of that product. Generators form the anchors throughout the grid for adequate voltage and frequency, have control systems to detect abnormal voltage or frequency, and quickly adjust when an anomaly is detected. In contrast, loads are the consumers of generation and generally do not take as active a role in assuring adequate voltages and frequency as generation does.

The CPUC acknowledges that when a large load is located and served from the high voltage transmission system, the ability of the grid to deliver power to customers could be impacted by creating incremental transmission congestion at the large load’s point-of-delivery. Further, if issues of deliverability and transfer capability are not addressed, there is the possibility that the wholesale generation market, and ratepayers,

⁴² *Id.*

could be impacted. A mechanism to retain or replace the transmission system's overall deliverability via robust and widespread transfer capabilities needs to be protected or restored. Such protection or restoration is best assured through more localized regulatory oversight with federal resources in support.

Sharing the goals of the DOE and FERC of prompt and efficient interconnection of large loads to the transmission system, the CPUC appreciates the opportunity to submit these comments and believes that it is important for the energization of large retail loads to remain under the purview of the states, as they are best situated to address local operational, reliability, and affordability concerns. The CPUC encourages the Commission to conduct this rulemaking in a way that embraces cooperative federalism, acknowledges the jurisdictional authority of the states, and helps to support the efforts underway in many states throughout the country engaging in rulemaking just like this one to ensure timely and efficient energizations, while protecting reliability, resource adequacy, and affordability for all ratepayers.

November 21, 2025

Respectfully submitted,
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CERTIFICATE OF SERVICE

I hereby certify that I have on this day caused a copy of the foregoing document to be served upon all parties designated on the official service list in these proceedings.

Dated at San Francisco, California, this **21st** day of **November 2025**.

/s/ INNA VINOGRADOV
Inna Vinogradov